

Assigned Data Analysis Project

1. Introduction

This study investigates the longitudinal progression of macular structural changes in glaucomatous eyes, focusing on Ganglion Cell Complex (GCC) thickness, a key indicator of retinal health. The study aims to characterize population-level trends in GCC thinning, evaluate individual variability, and assess whether baseline GCC thickness and age influence progression rates.

2. Exploratory Data Analysis

Over four years, 786 measurements were collected biannually, revealing significant baseline GCC differences between groups as shown in Table 1: $95.06 \pm 11.03 \mu\text{m}$ in the "High GCC" group and $67.35 \pm 6.39 \mu\text{m}$ in the "Low GCC" group ($p < 0.001$). Baseline age was also lower in the "High GCC" group (64.26 ± 8.68 years) compared to the "Low GCC" group (67.50 ± 7.84 years) ($p < 0.001$) as shown in Table 2. Figure 1 (profile plots) illustrates faster GCC declines in the "High GCC" group, while Figure 2 confirms a population-level decline of approximately $-0.79 \mu\text{m}/\text{year}$. Correlation matrices show strong correlations between closely spaced measurements, supporting structured covariance modeling. Mixed-effects modeling indicates baseline GCC significantly interacts with time ($p = 0.0355$), suggesting faster thinning for patients with higher baseline GCC.

3. Population time trend & best fitting covariance model.

The quadratic model was determined to provide the best fit for GCC thinning trends, as it had the lowest Akaike Information Criterion (AIC) value (4068.885), compared to the linear model (4082.154) and cubic model (4069.448) (Table 3). The quadratic model captures subtle

curvature in the population trend, offering a better balance between complexity and explanatory power than the linear model, while avoiding overfitting associated with the cubic model. Figure 3 visualizes the population-level trend, showing a consistent decline in GCC thickness over time with minimal non-linear variation. The fitted line aligns with the scatter points, emphasizing the adequacy of the model in explaining the observed data.

Eight covariance models were evaluated using AIC to identify the best-fitting structure (Table 4). The Random Intercept and Slope (RIAS) model, with an AIC of 4082.154, provided the best fit, accounting for population-level changes and individual variability. Temporal correlation models like ARMA(1,1) (4091.449) and ARMA(2,1) (4093.325) performed slightly worse. Compound Symmetry (CS) and Random Intercept Only (RI) models had the poorest fits (AIC > 4228). The RIAS model effectively balances complexity and fit, making it the optimal choice for analyzing GCC progression.

4. Macular Thickness Loss Rates and Baseline GCC Influence

The population average rate of GCC loss is -1.302 microns per year (Table 5), with individual rates ranging from -6.194 to 2.353 microns per year and a standard deviation of 1.283 microns, indicating substantial variability among patients. Most rates cluster near the mean, but some individuals exhibit significantly faster or slower progression (Figure 4), underscoring the need for personalized monitoring.

Regression analysis (Table 6) shows a weak negative relationship between baseline GCC thickness and the rate of loss (slope = -0.0096, $p = 0.191$), suggesting patients with higher initial GCC may thin slightly faster. However, the lack of significance indicates baseline GCC is not a strong predictor, and other factors likely influence progression. Figure 5

illustrates this weak trend, with confidence intervals overlapping zero, highlighting the limited evidence for this hypothesis.

5. Age Effect

The analysis examined the role of age in predicting GCC thickness and its trends in glaucomatous patients, using a population intercept and slope model with a random intercept and slope for the covariance structure. Two models were compared: a base model with time as the sole predictor and an age-adjusted model that included `base_age`. The age-adjusted model had a slightly higher AIC (4083.455) than the base model (4082.154), indicating that including `base_age` did not improve the model's fit. This suggests that age does not significantly contribute to explaining variation in GCC trends across patients.

Table 7 for the age-adjusted model reinforces this conclusion. The effect of `base_age` on GCC thickness trends was small (-0.281 microns per year of age) and not statistically significant ($p = 0.141$). While older patients might have slightly thinner GCC on average, this effect is too weak to conclude that age meaningfully influences GCC progression.

A visualization (Figure 6) of GCC thickness trends over time, grouped by `base_age`, highlights substantial variability among individuals but reveals no consistent relationship between age and GCC thinning. This lack of a clear pattern aligns with the statistical results, supporting the conclusion that age is not a key factor in GCC trends.

Overall, the analysis shows that GCC thickness decreases significantly over time, with age having little impact on this trend. Other factors, such as disease severity or treatment history, may better explain GCC progression. The use of a population and random slope model effectively captured individual variability, underscoring the limited role of age in this dataset.

Appendix

Table 1

Characteristics of GCC Groups					
Group	n	GCC	Base Age	GCC p [†]	Age p [†]
High GCC	399	95.06 ± 11.03	64.26 ± 8.68	<0.001	<0.001
Low GCC	387	67.35 ± 6.39	67.5 ± 7.84		
[†] p-values from t-tests.					
Table 1					

Table 2

GCC_group <chr>	Mean_Age <dbl>	SD_Age <dbl>	n <int>
High GCC	64.25815	8.678283	399
Low GCC	67.50388	7.836864	387

Table 3

Model <chr>	AIC <dbl>
Linear	4082.154
Quadratic	4068.885
Cubic	4069.448

Table 4

	Model <chr>	AIC <dbl>
RIAS	RIAS	4082.154
ARMA11	ARMA11	4091.449
ARMA21	ARMA21	4093.325
ARMA20	ARMA20	4100.158
AR1	AR1	4115.166
TOEP	TOEP	4115.166
RI	RI	4228.208
CS	CS	4230.208

Table 5

Mean <dbl>	SD <dbl>	Min <dbl>	Max <dbl>
-0.8206987	0.9795553	-4.482591	1.219852

Table 6

```

Call:
lm(formula = individual_slope ~ initial_GCC, data = individual_slopes)

Residuals:
    Min       1Q   Median       3Q      Max
-5.2759 -0.4583  0.1764  0.7149  3.3583

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.022933   0.626021  -0.037   0.971
initial_GCC -0.009627   0.007322  -1.315   0.191

Residual standard error: 1.278 on 105 degrees of freedom
Multiple R-squared:  0.01619,    Adjusted R-squared:  0.006825
F-statistic: 1.728 on 1 and 105 DF,  p-value: 0.1915

```

Table 7

	Value <chr>	Std.Error <chr>	DF <chr>	t-value <chr>	p-value <chr>
(Intercept)	101.82166	12.628819	678	8.062643	0.0000
time	-0.82092	0.122425	678	-6.705495	0.0000
base_age	-0.28108	0.189711	105	-1.481599	0.1414

Figure 1

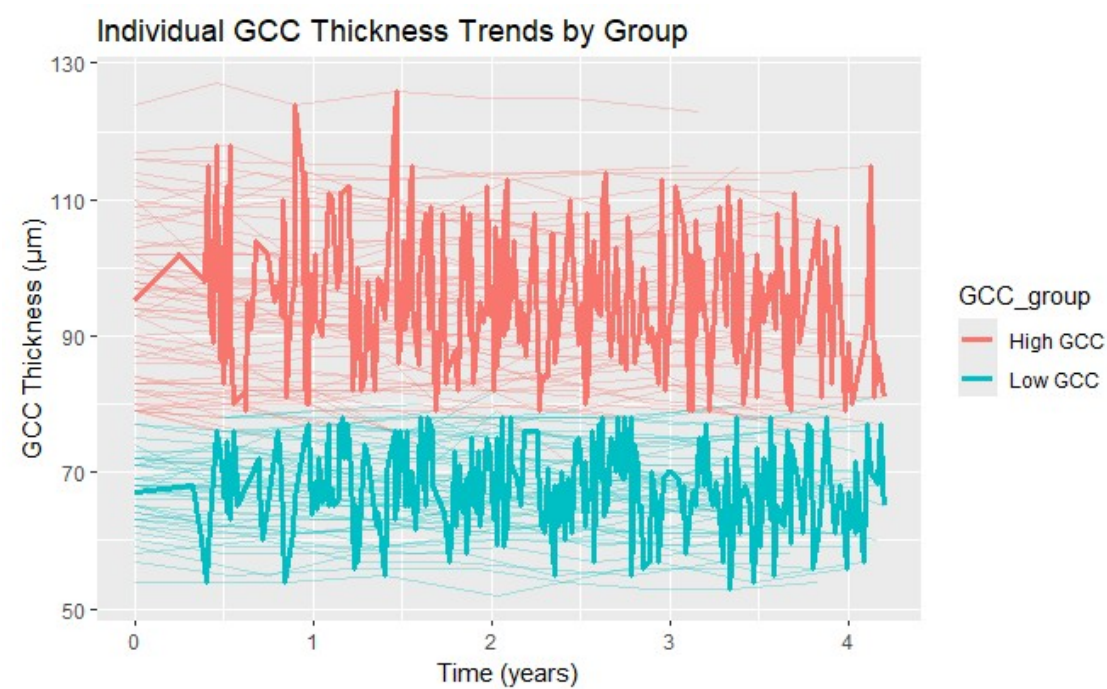


Figure 2

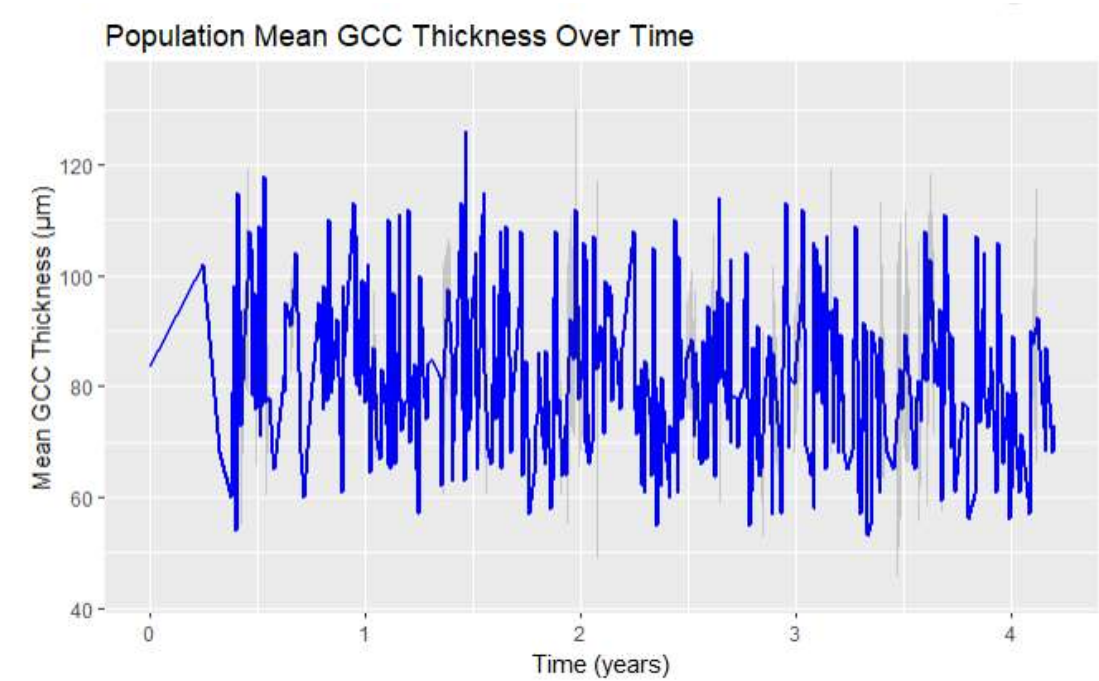


Figure 3

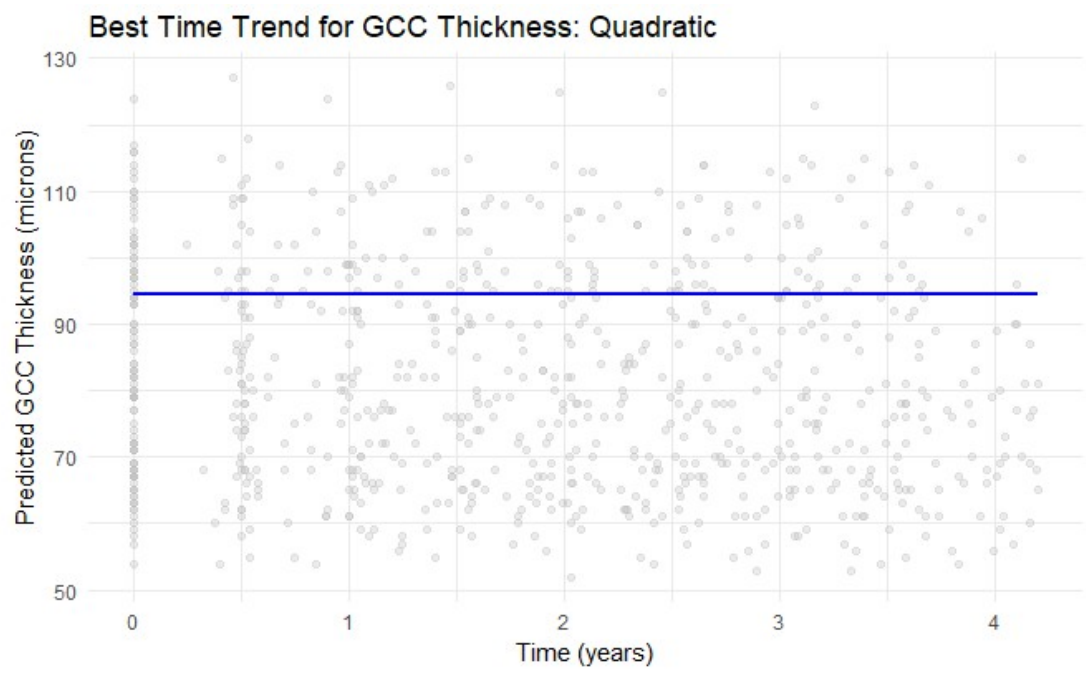


Figure 4

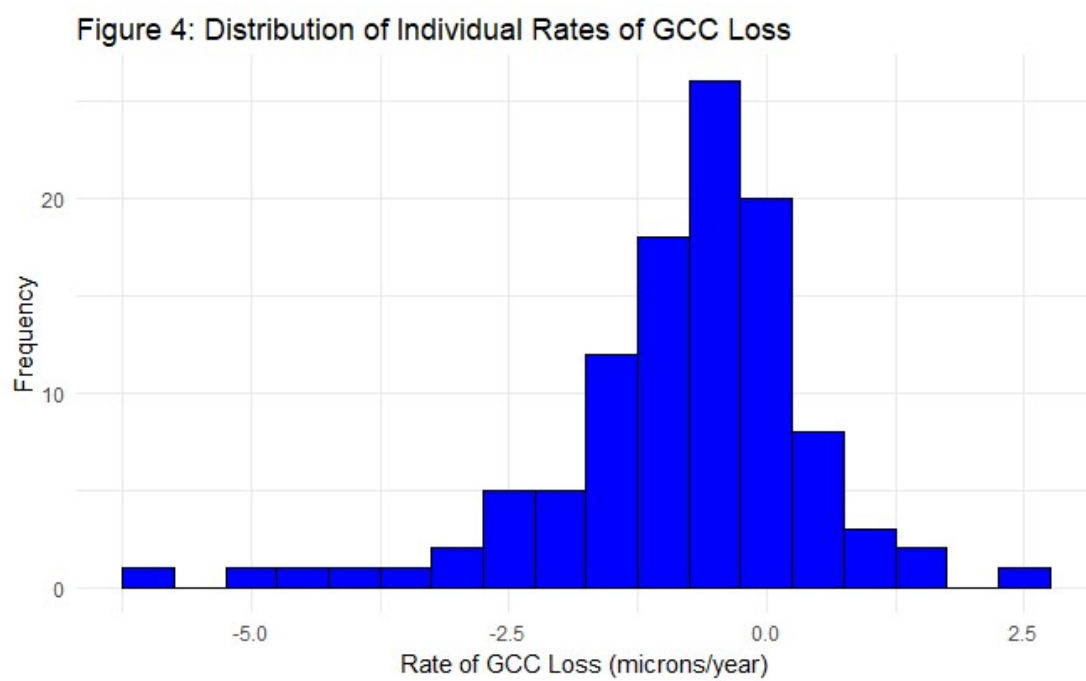


Figure 5

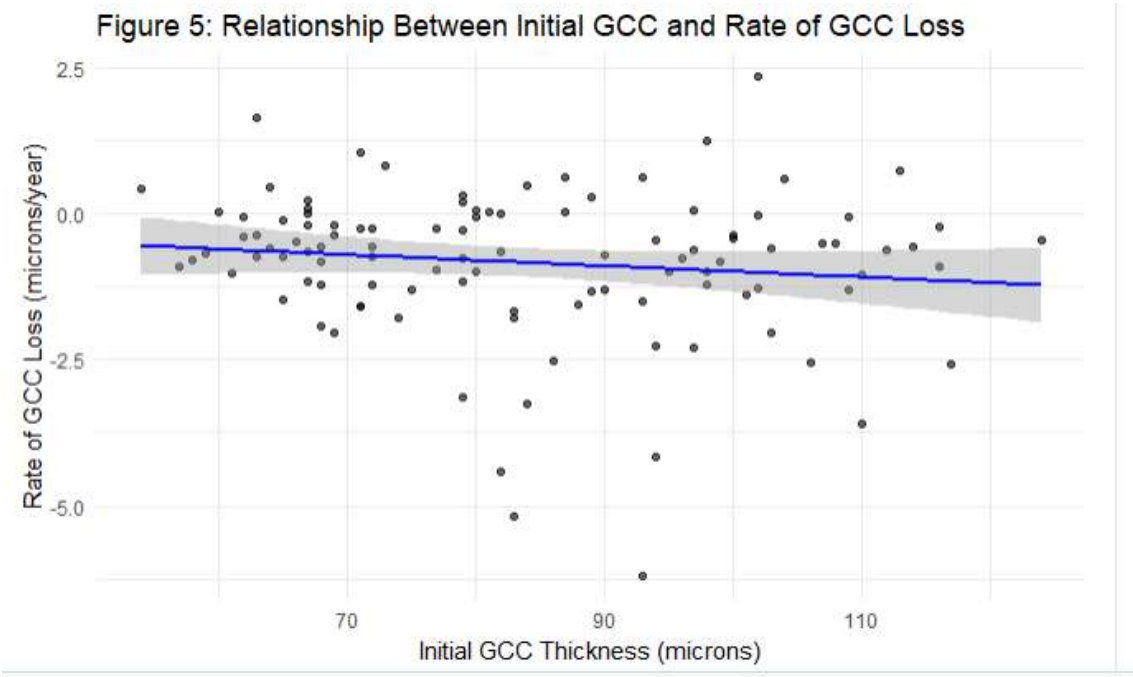


Figure 6

